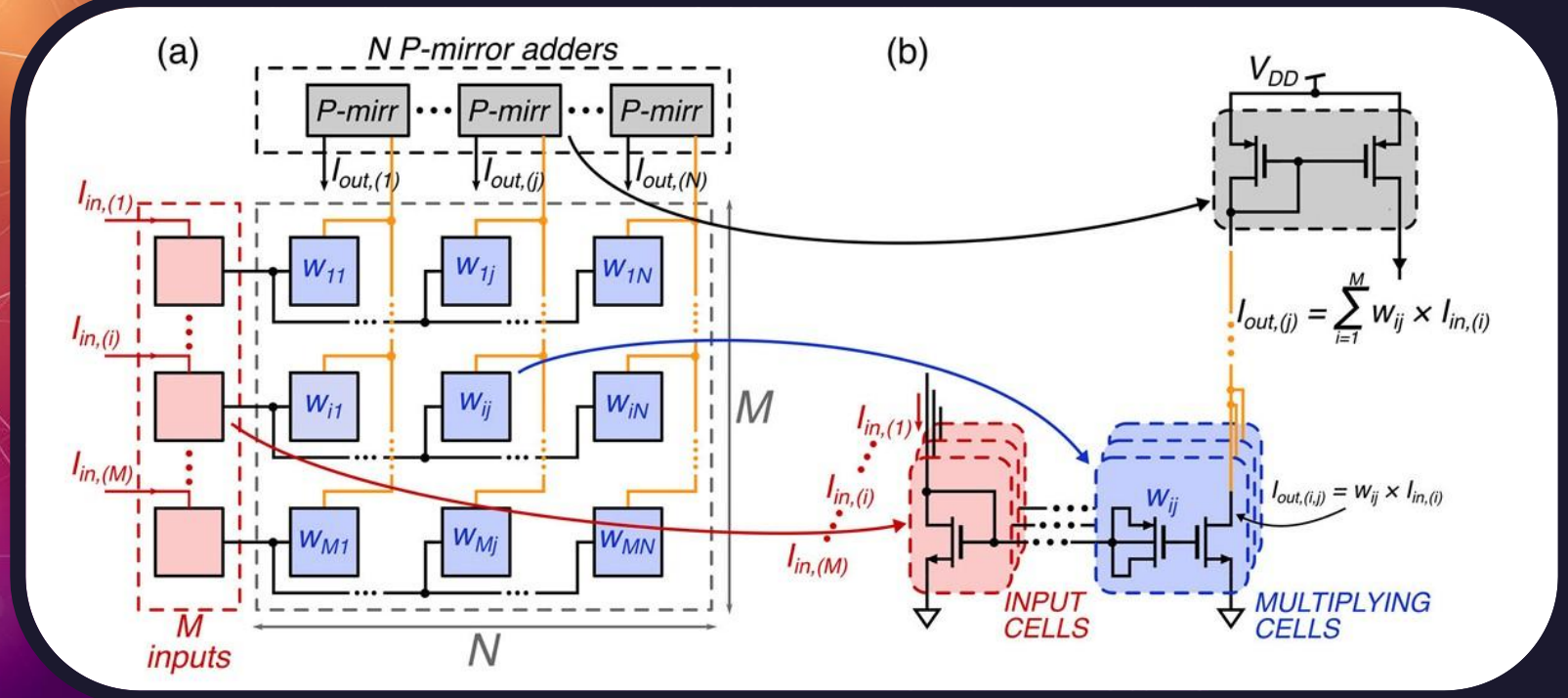
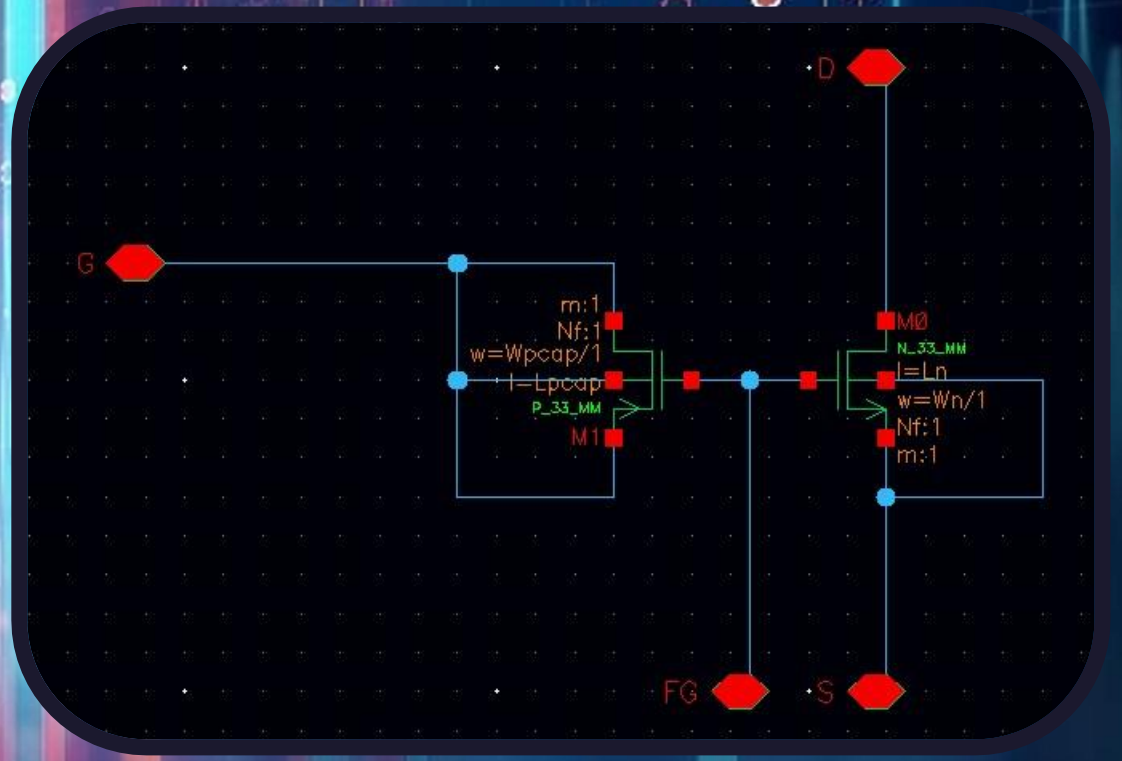
Analog VMM analysis

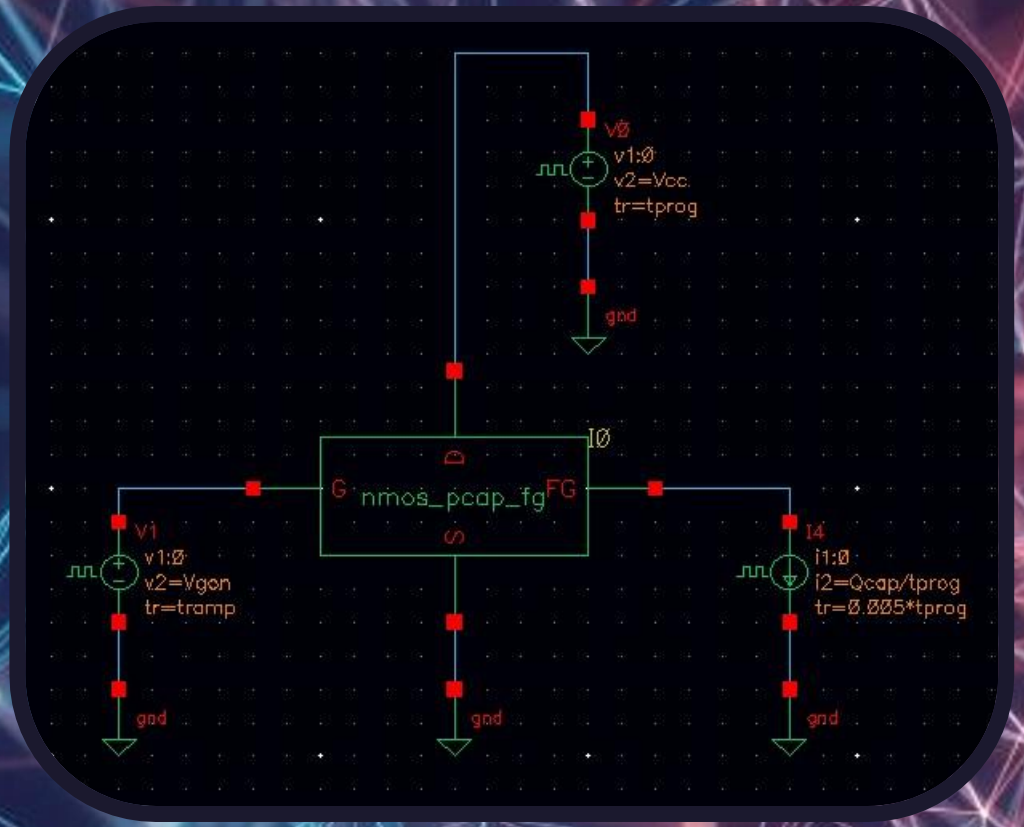


Floating gate cell design

The floating-gate cell used is made up of an n-mos with a p-cap connected in series with the gate.

The value of the voltage stored in the p-cap, being a drop in series with the mos gate, has the same effect as an increase in the threshold voltage (Vth).





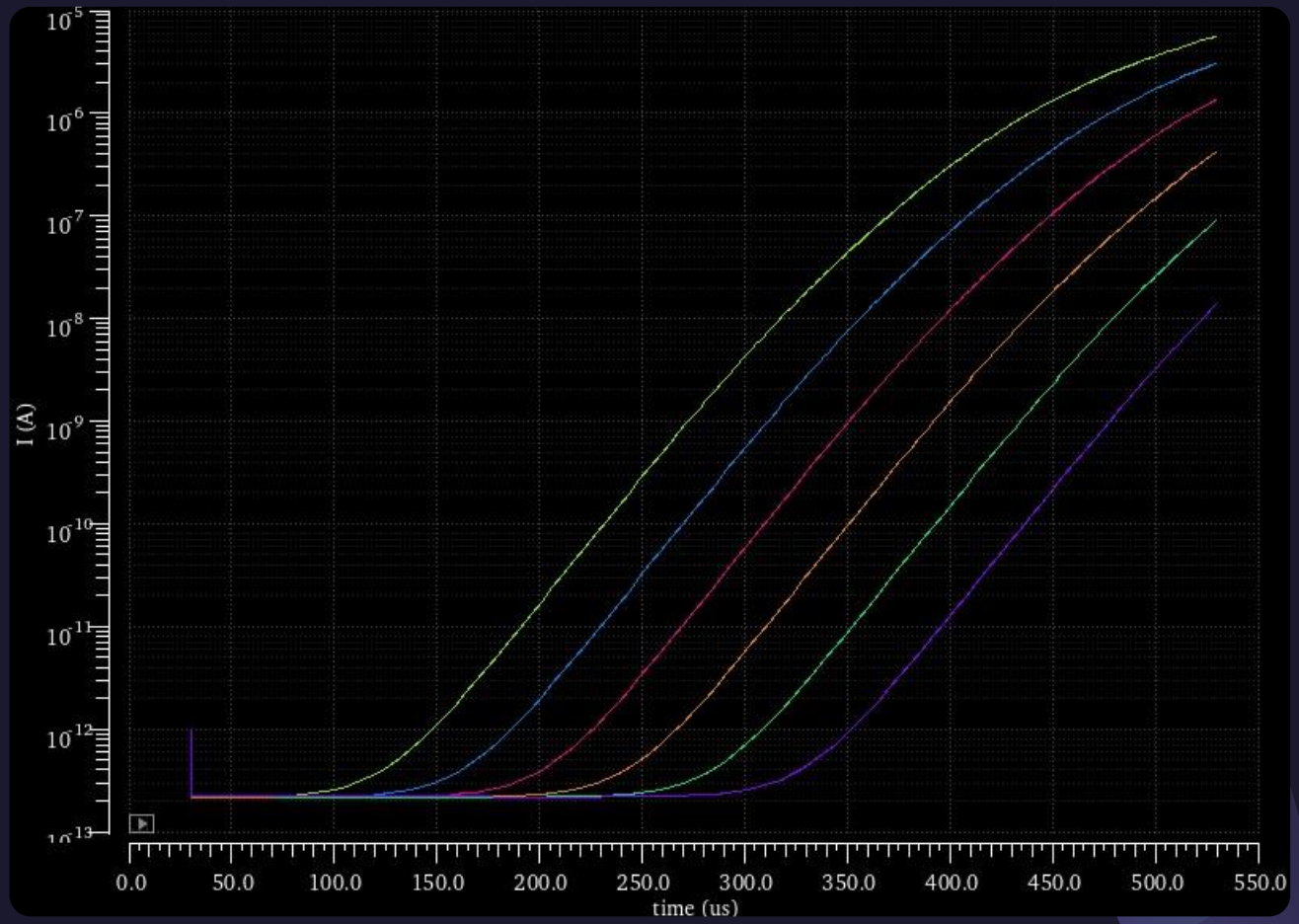
Weak inversion polarization of the n-mos cell

The single n-mos cell must work in weak inversion, for this reason the drain current have to be limited to a maximum value calculated as follows.

$$\frac{I_{ds}}{I_{dspec}} \leq \frac{1}{10} \rightarrow I_{ds} \leq \frac{I_{dspec}}{10} = \frac{I'_{dspec}\frac{W}{L}}{10}$$

Were $I'_{dspec} \cong 0{,}2\mu A$ and W/L is the aspect ratio of the n-mos.

With the simulation we want to verify that with the calculated drain current, the n-mos work in weak inversion.

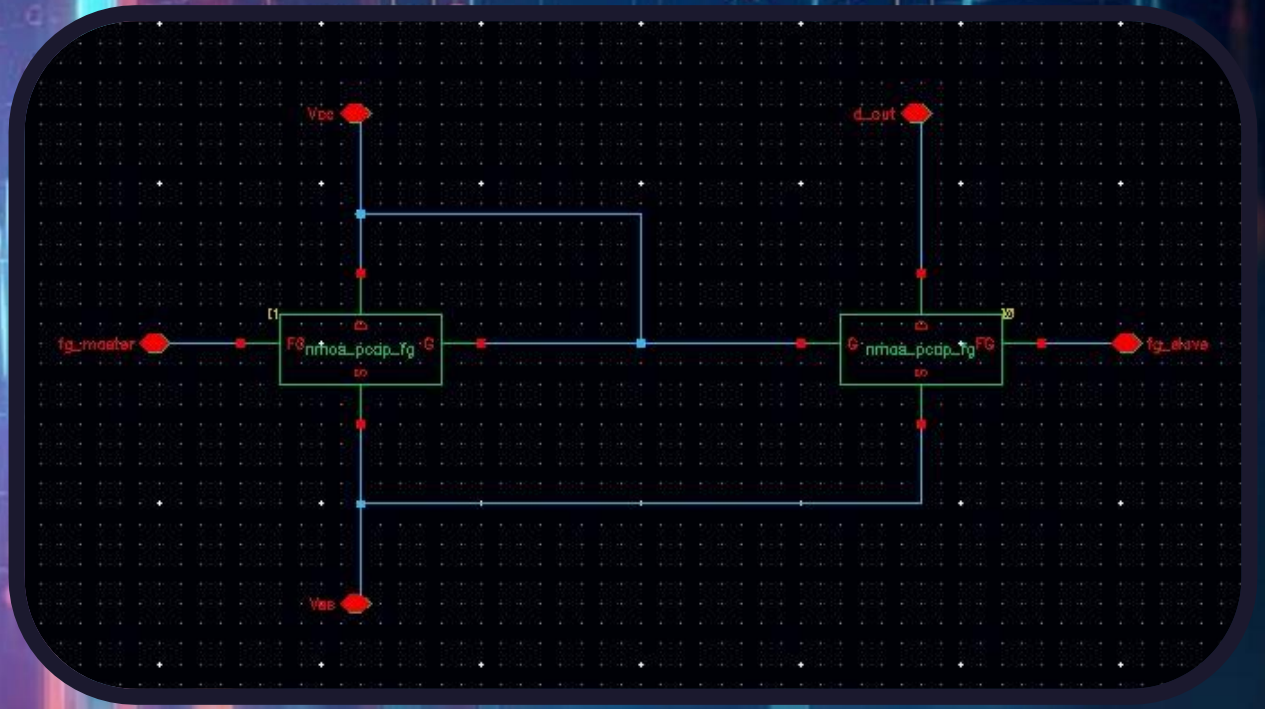


The simulation is performed by charging the p-cap with different charge values and applying a voltage ramp to the gate of the n-mos, while measuring the drain current.

- The gate voltage goes from 0V to 1V
- The p-cap charge goes from 1fC to 20fC with 6 steps
- The n-mos aspect ratio is Wn/Ln=1um/0,8um
- The drain source voltage is Vds=1V

As we can see from the graph, the curve are linear when the drain current is under approximately $20nA$.

With the n-mos dimension analysed the maximum drain current is $I_{ds} \leq \frac{I'_{dspec}\frac{W}{L}}{10} \cong 25nA$



Current mirror cell design

Two floating gate cells are used to make a current mirror.

The voltage drop on the p-cap produce a shift of the threshold voltage determining the current magnification factor of the mirror.

The n-mos are in weak inversion, so the output current is calculated as follows

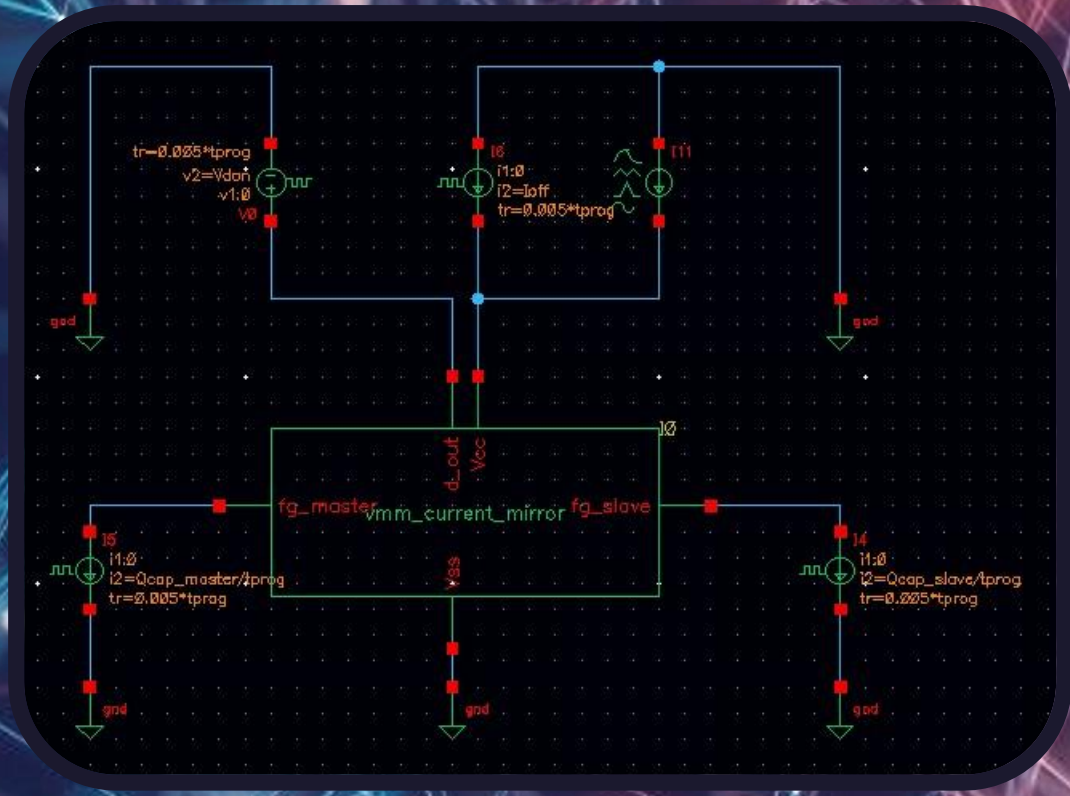
$$I_{out} = I_{in} \cdot e^{\frac{\Delta V_{th}}{mV_t}}$$

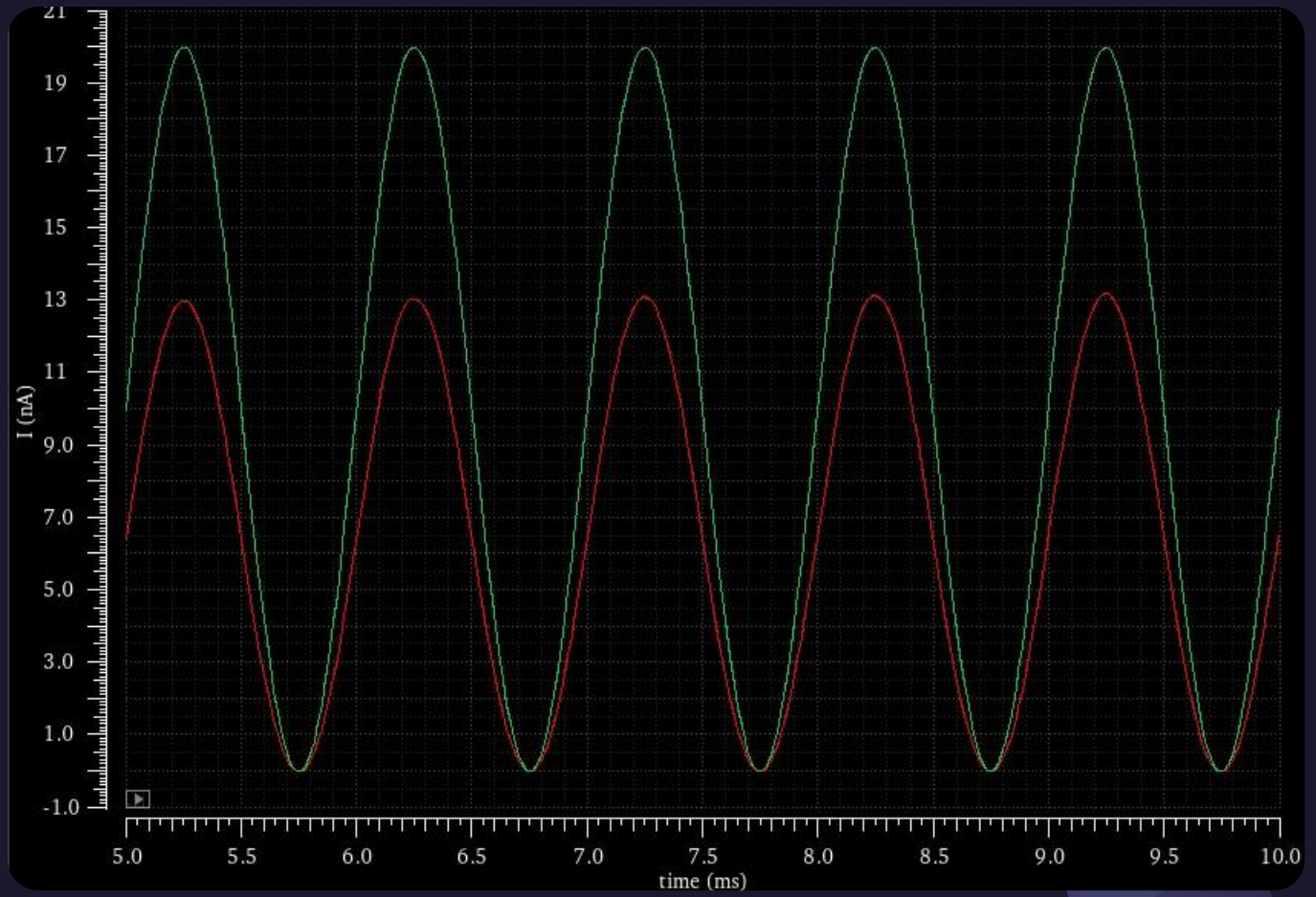
Magnification factor analysis for the current mirror

The output current value of the current mirror is calculated as follows.

$$\begin{cases} I_{out} = w \cdot I_{in} \\ w = e^{\frac{\Delta V_{th}}{m V_t}} \end{cases}$$

The magnification factor was evaluated by charging the p-cap with different charge levels and measuring the mirror's output current with a sinusoidal input current applied.





- The green sinusoidal wave is the input current that goes into the master transistor
- The red sinusoidal wave is the output current from the slave transistor

The dimensions for the n-mos and for the p-cap are

- Wn/Ln = 1um/0,8um
- Wpcap*Lpcap = 5um*5um

The p-cap is charged with 1fC and the drain voltage is Vds=600mV

The magnification factor from the analysis is

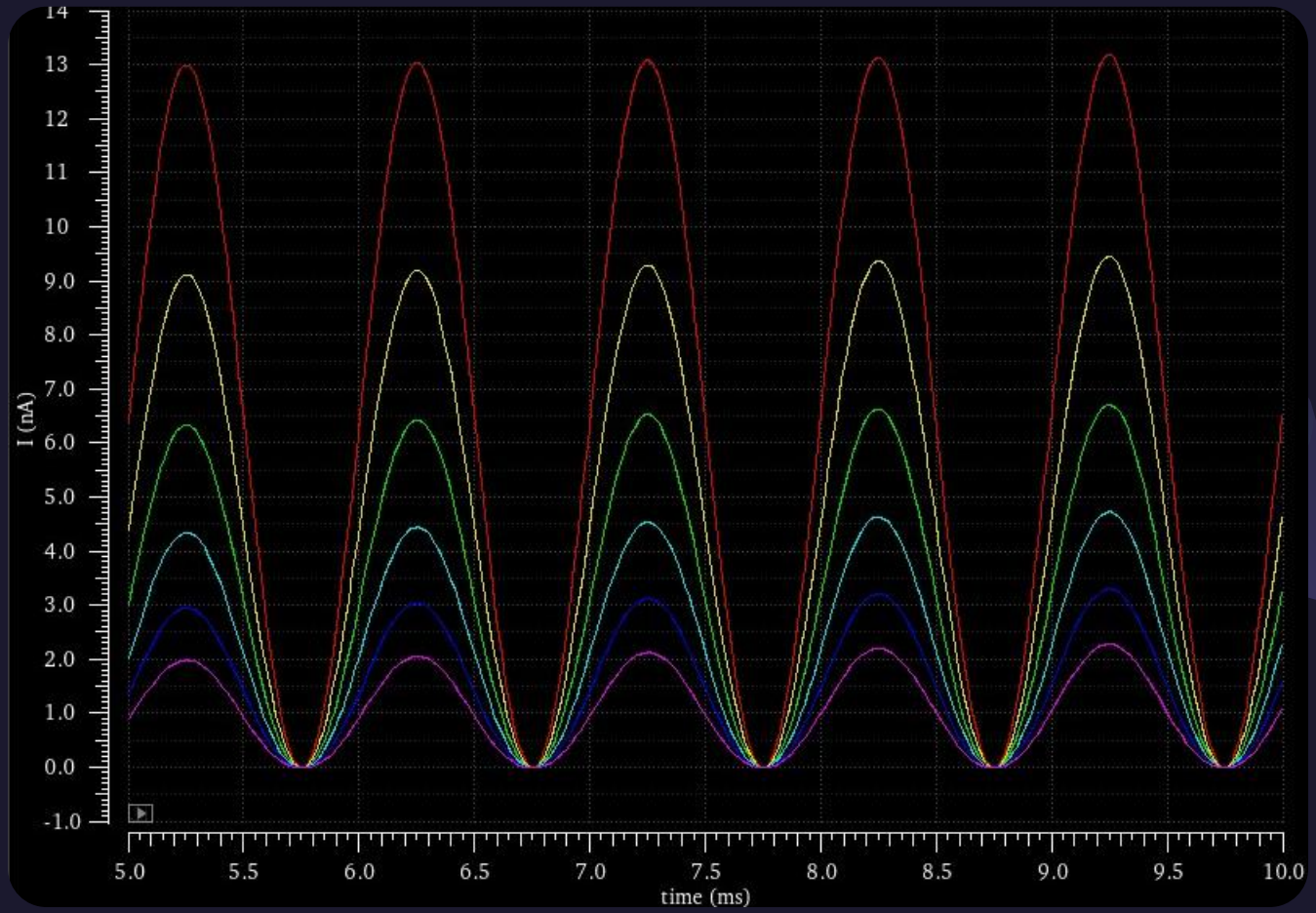
$$w = \frac{I_{slave}}{I_{master}} \cong 0{,}65$$

The theoretical calculated value is

$$w = e^{\frac{\Delta V_{th}}{mV_t}} = e^{\frac{-Q_{cap}}{mV_t \cdot W_{cap} L_{cap} C_{ox}}} \cong 0{,}60$$

After the first analysis we applied different value of charge at the p-cap to see the variation of magnification value

The charge value applied goes from 1fC to 5fC with a step size of 0,8fC (resulting in 6 steps)



The magnification values corresponding to each curve are the following

- Red curve ($Q_{cap} = 1fC$)

  $w_{simulated} \cong 0{,}65$; $w_{calculated} \cong 0{,}60$

- Yellow curve ($Q_{cap} = 1{,}8fC$)

  $w_{simulated} \cong 0{,}45$; $w_{calculated} \cong 0{,}40$

- Green curve ($Q_{cap} = 2{,}6fC$)

  $w_{simulated} \cong 0{,}31$; $w_{calculated} \cong 0{,}26$

- Light blue curve ($Q_{cap} = 3{,}4fC$)

  $w_{simulated} \cong 0{,}21$; $w_{calculated} \cong 0{,}18$

- Blue curve ($Q_{cap} = 4{,}2fC$)

  $w_{simulated} \cong 0{,}15$; $w_{calculated} \cong 0{,}12$

- Purple curve ($Q_{cap} = 5fC$)

  $w_{simulated} \cong 0{,}10$; $w_{calculated} \cong 0{,}08$

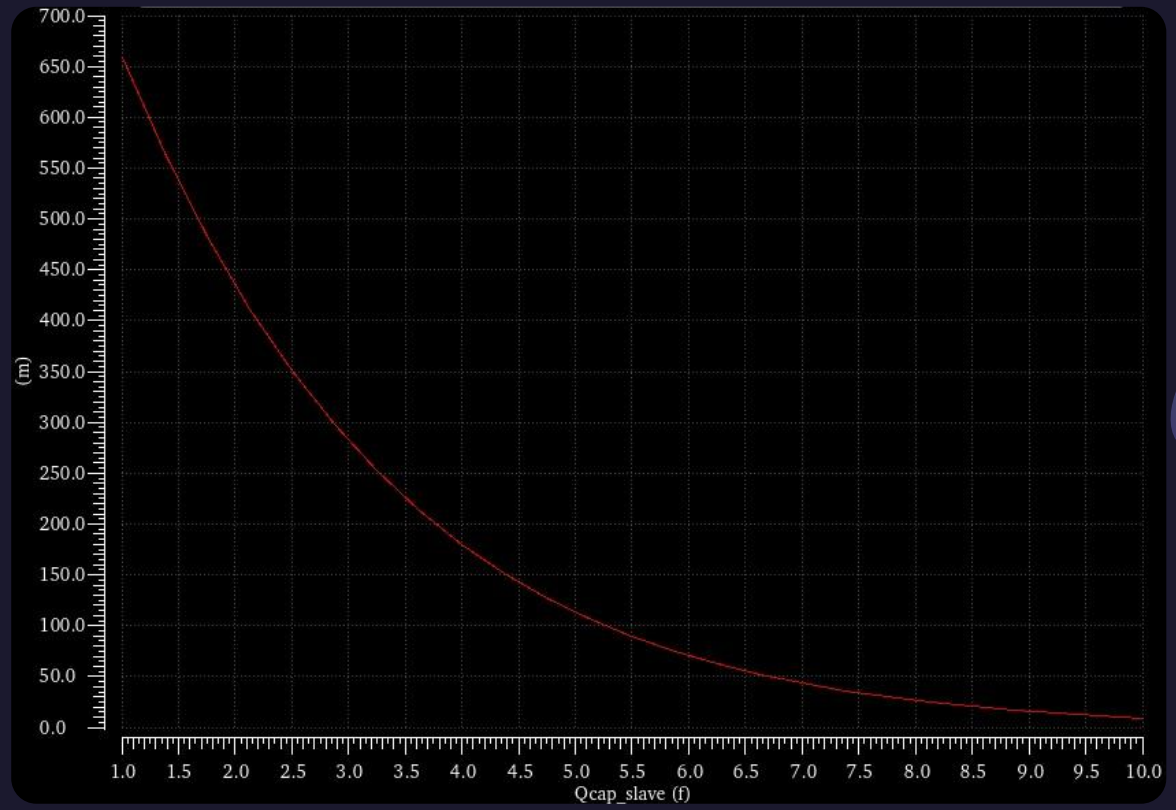
The magnification factor (equal to the weight of the cell) function to the charge into the p-cap is the following.

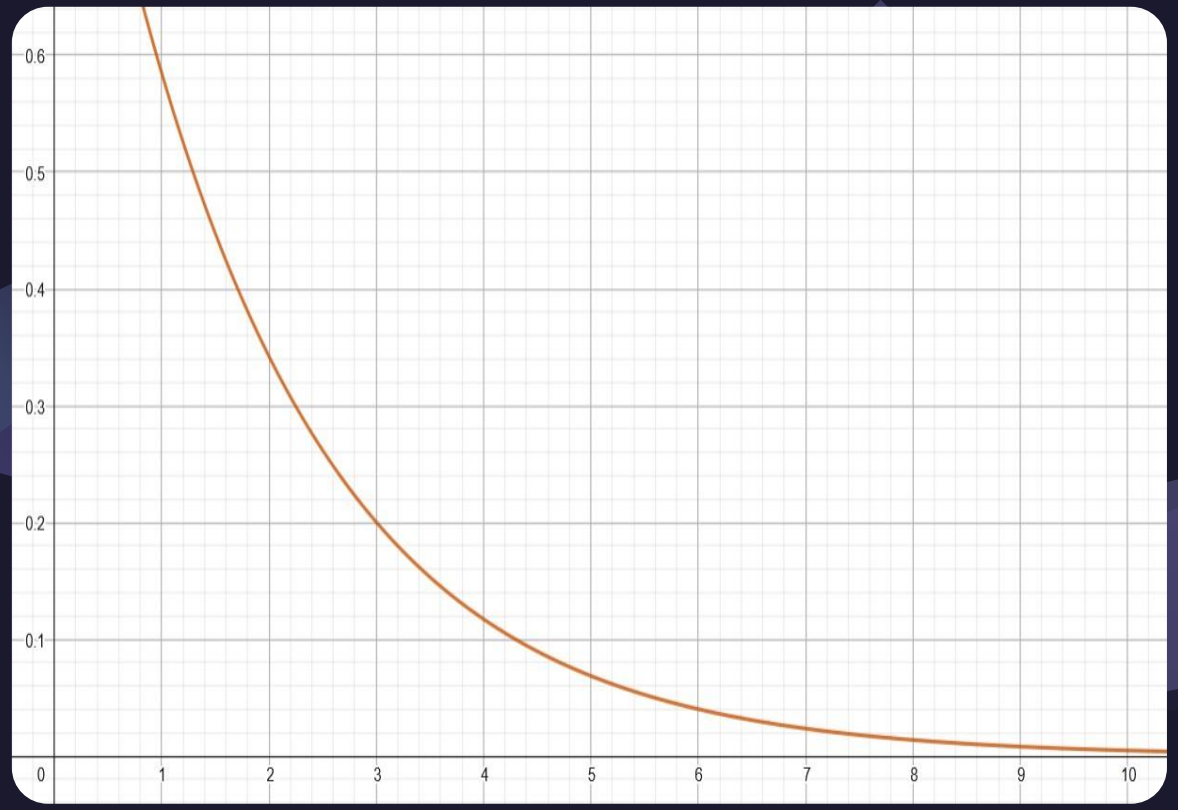
With a charge swing from 1fC to 10fC we can program the magnification factor from 0,65 to almost zero.

Comparing the result with the theoretical function calculated earlier

$$y(x) = e^{-\frac{x \cdot 10^{-15}}{1{,}866 \cdot 10^{-15}}}$$

the result follows the model pretty well.



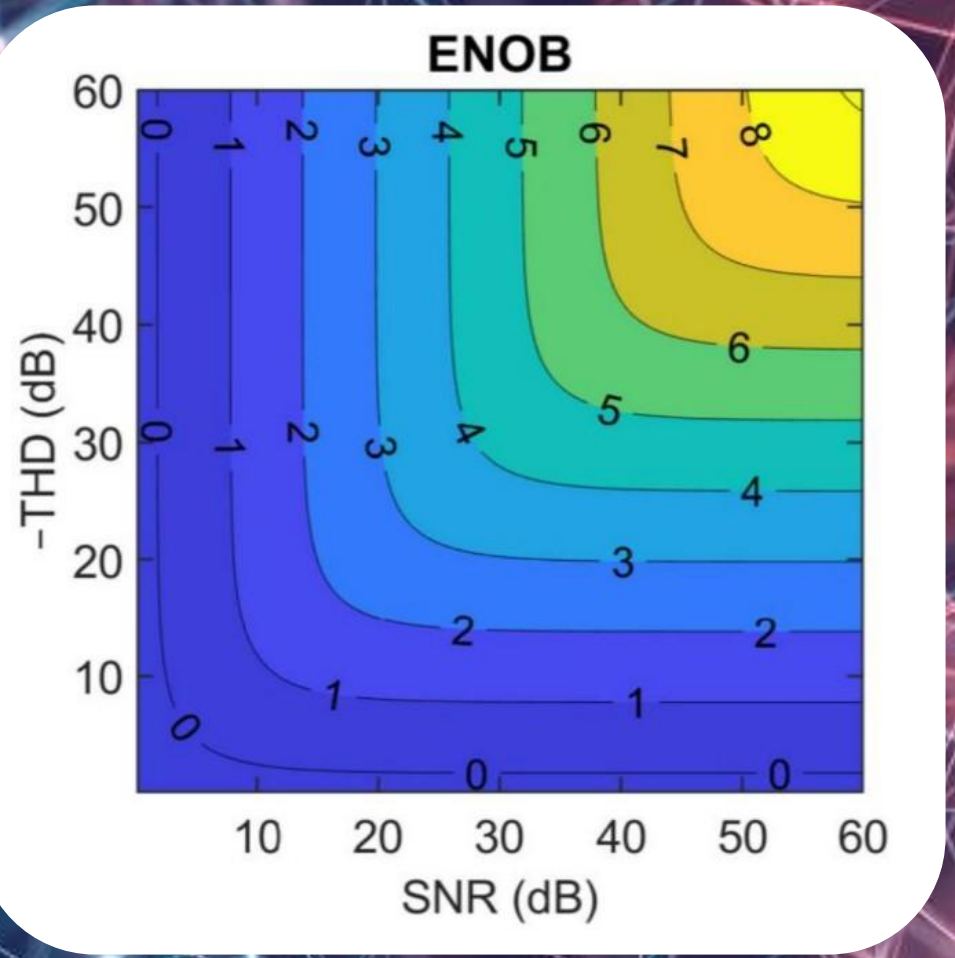


Spectrum analysis

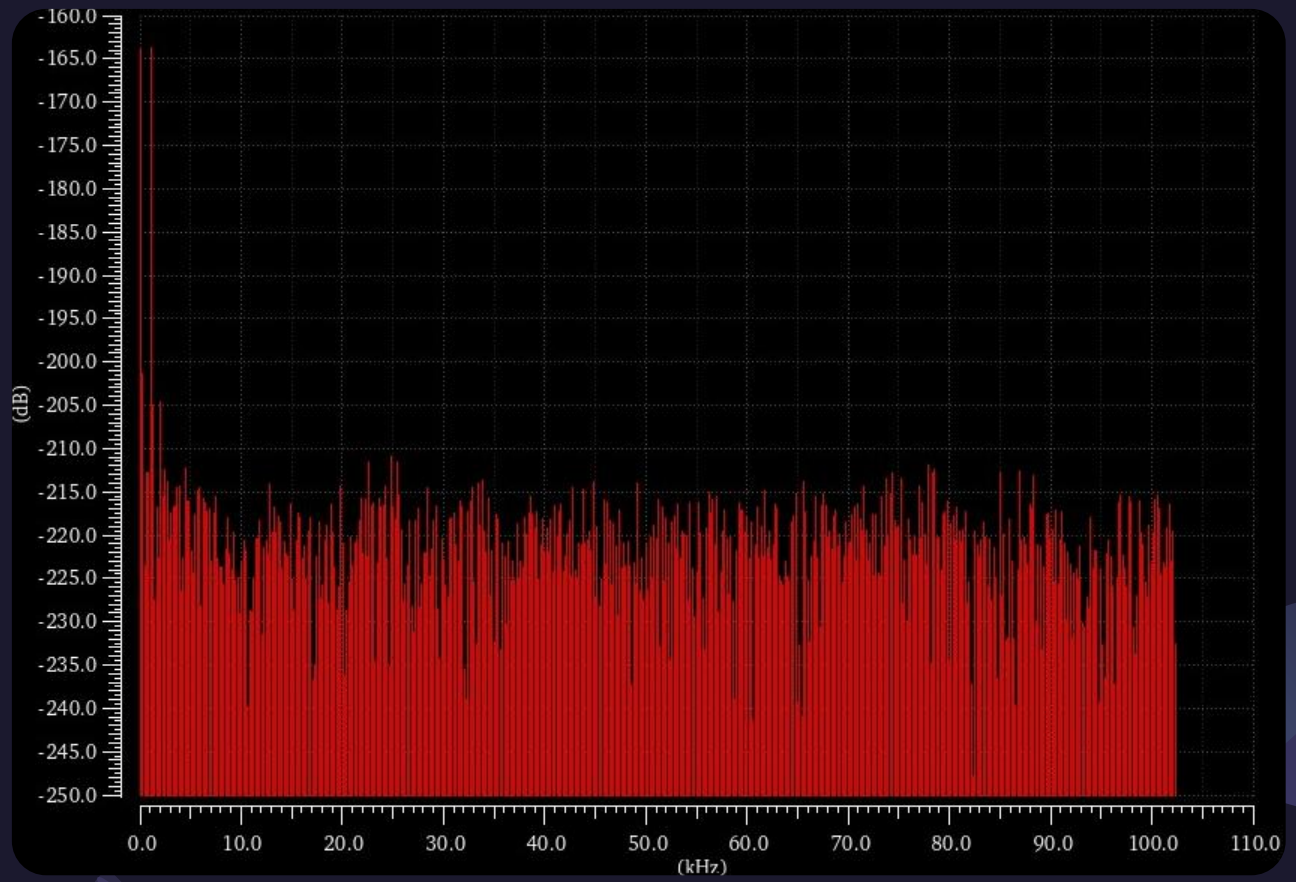
With the spectrum analysis, we apply a pure sine-wave as input and see the spectrum of the mirrored current signal.

With the spectrum analysis, the simulator can calculate these FOMs

- $SNR = 10 \cdot \log(\frac{I_{signal}^2}{I_{noise}^2})$
- $THD = -10 \cdot \log(\frac{I_{signal}^2}{I_{distorsion}^2})$
- $SINAD = 10 \cdot \log(\frac{I_{signal}^2}{I_{noise}^2 + I_{distorsion}^2})$
- $ENOB = \frac{SINAD - 1{,}76}{6{,}02}$



Spectrum analysis



This is the spectrum of the mirrored output wave considering the transient noise of the components.

As we can see, there are two main frequency components that are higher that the others

- The component at $f = 0Hz$ that is present because the sine-wave is not a zero-mean signal
- The component at $f = 1kHz$ that is the fundamental harmonic of the output signal

The spectrum analysis has been done with n-mos dimensions of Wn/Ln = 1um/0,8um and p-cap dimension of Wpcap*Lpcap = 5um*5um

In this case the p-cap is charged, so the current mirror have a magnification factor (value of w) different from 1. The calculated $ENOB$ refers to a signal that have a magnification factor equal to 1, so the value needs to be corrected by adding the value $-\log_2 w$

Dependence of ENOB on maximum input current amplitude

The amplitude of the noise frequency components (thermal noise) is independent of the amplitude of the input current waveform.

This means that I^2_{signal} decrease. The $I^2_{distorsion}$ also decrease, while I^2_{noise} remains the same.

As a result, the SNR increase and THD decreases, this means that $SINAD$ ad $ENOB$ increases with the input current amplitude.

We have simulated these values for three different values of input current amplitude to make a comparison. The simulation has been done with Vds = 1V, without charging the p-cap, with n-mos dimensions of Wn/Ln = 1um/0,8um and p-cap dimension of Wpcap*Lpcap = 5um*5um.

- Maximum input current amplitude of 28nA

$ENOB \cong 5{,}49bits$

$SNR \cong 34{,}99dB; THD \cong -38{,}74dB$

- Maximum input current amplitude of 10nA

$ENOB \cong 5{,}06bits$

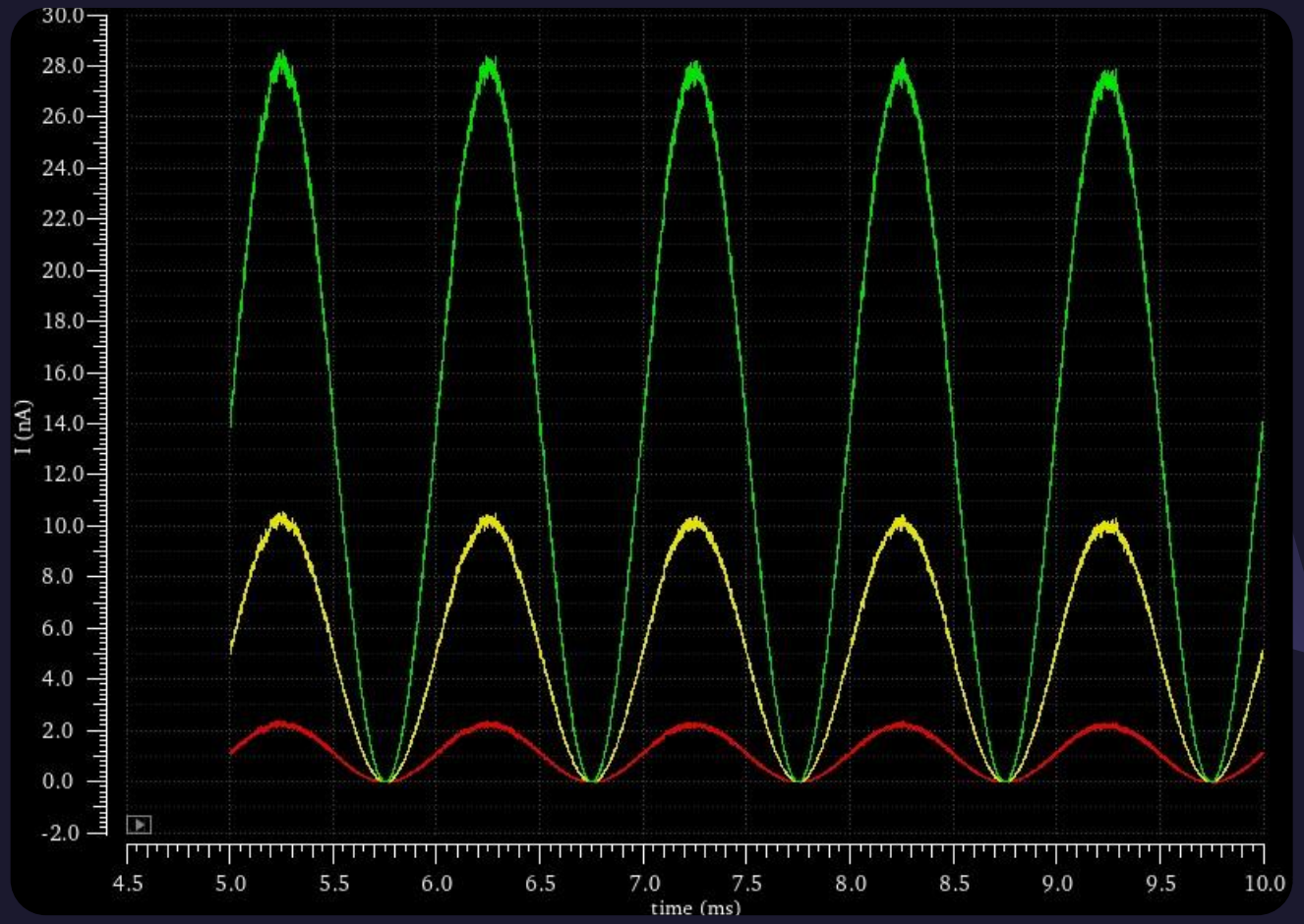
$SNR \cong 33{,}63dB; THD \cong -39{,}27dB$

- Maximum input current amplitude of 2nA

$ENOB \cong 4{,}39bits$

$SNR \cong 29{,}64dB; THD \cong -39{,}90dB$

Dependence of ENOB on maximum input current amplitude



- Maximum input current amplitude of 28nA

$ENOB \cong 5{,}49bits$

$SNR \cong 34{,}99dB; THD \cong -38{,}74dB$

- Maximum input current amplitude of 10nA

$ENOB \cong 5{,}06bits$

$SNR \cong 33{,}63dB; THD \cong -39{,}27dB$

- Maximum input current amplitude of 2nA

$ENOB \cong 4{,}39bits$

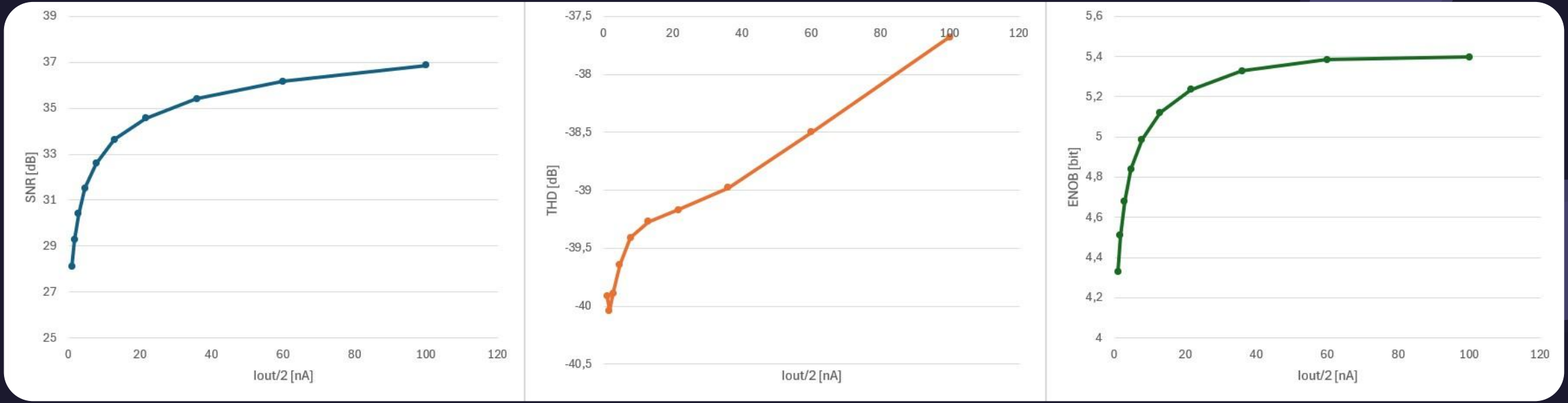
$SNR \cong 29{,}64dB; THD \cong -39{,}90dB$

Dependence of ENOB on maximum input current amplitude

As we can see from the graph, the SNR increases with the input current amplitude because the noise spectral component amplitudes are independent of the input current amplitude.

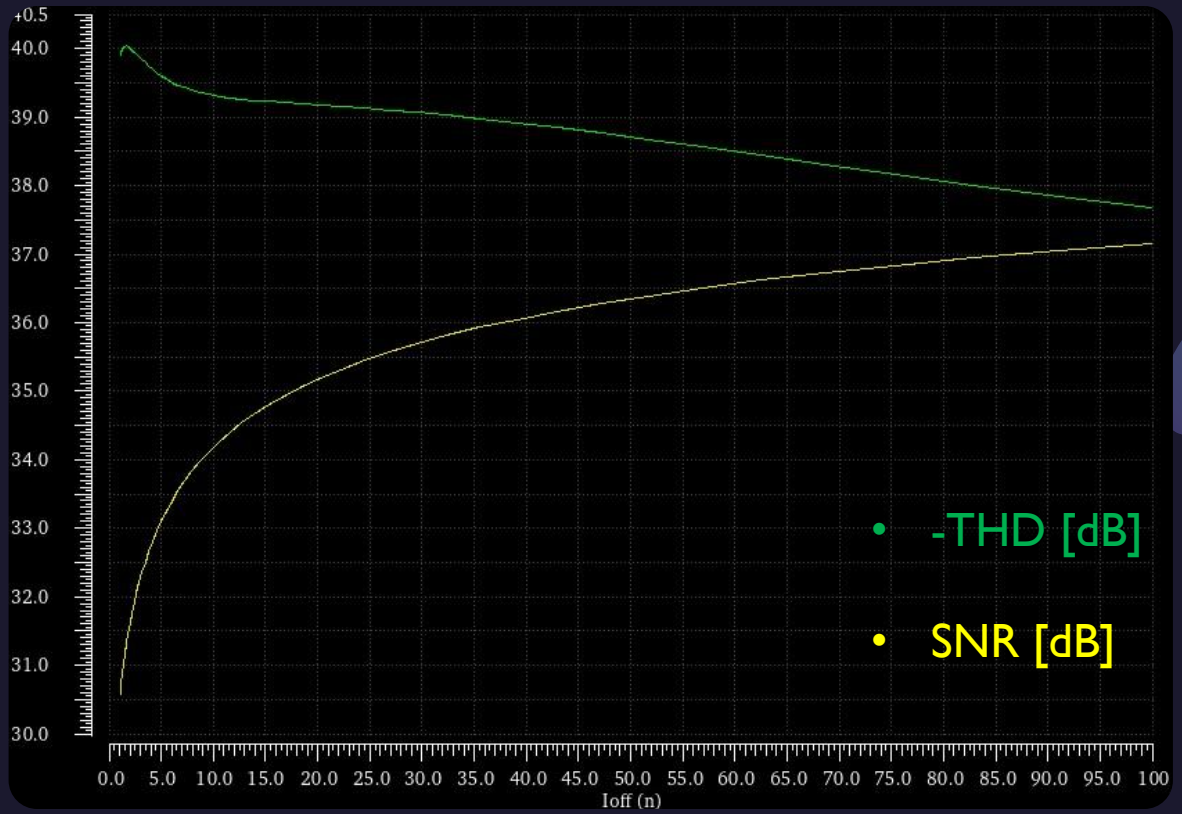
Moreover, the THD decreases because a smaller signal introduces fewer nonlinear effects.

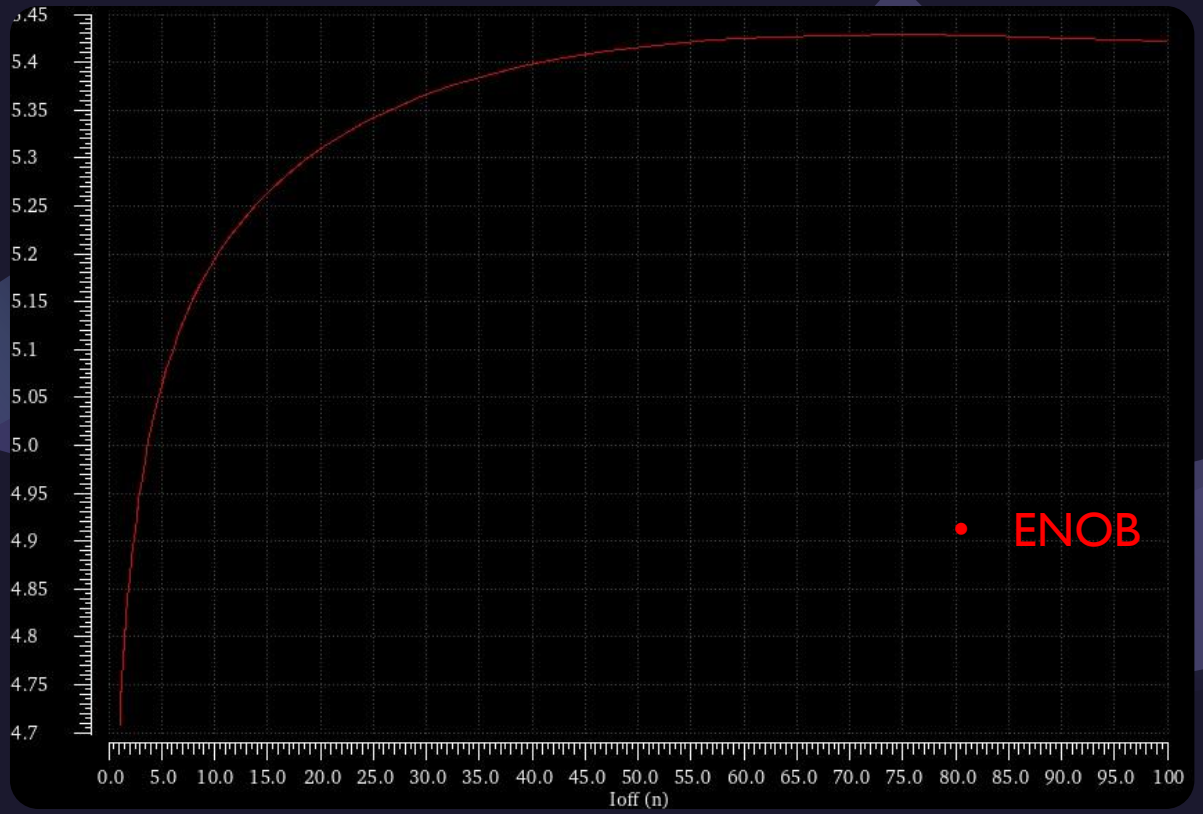
In the graph the current value is Iout/2 because the magnification factor is 1, so Iout = Iin



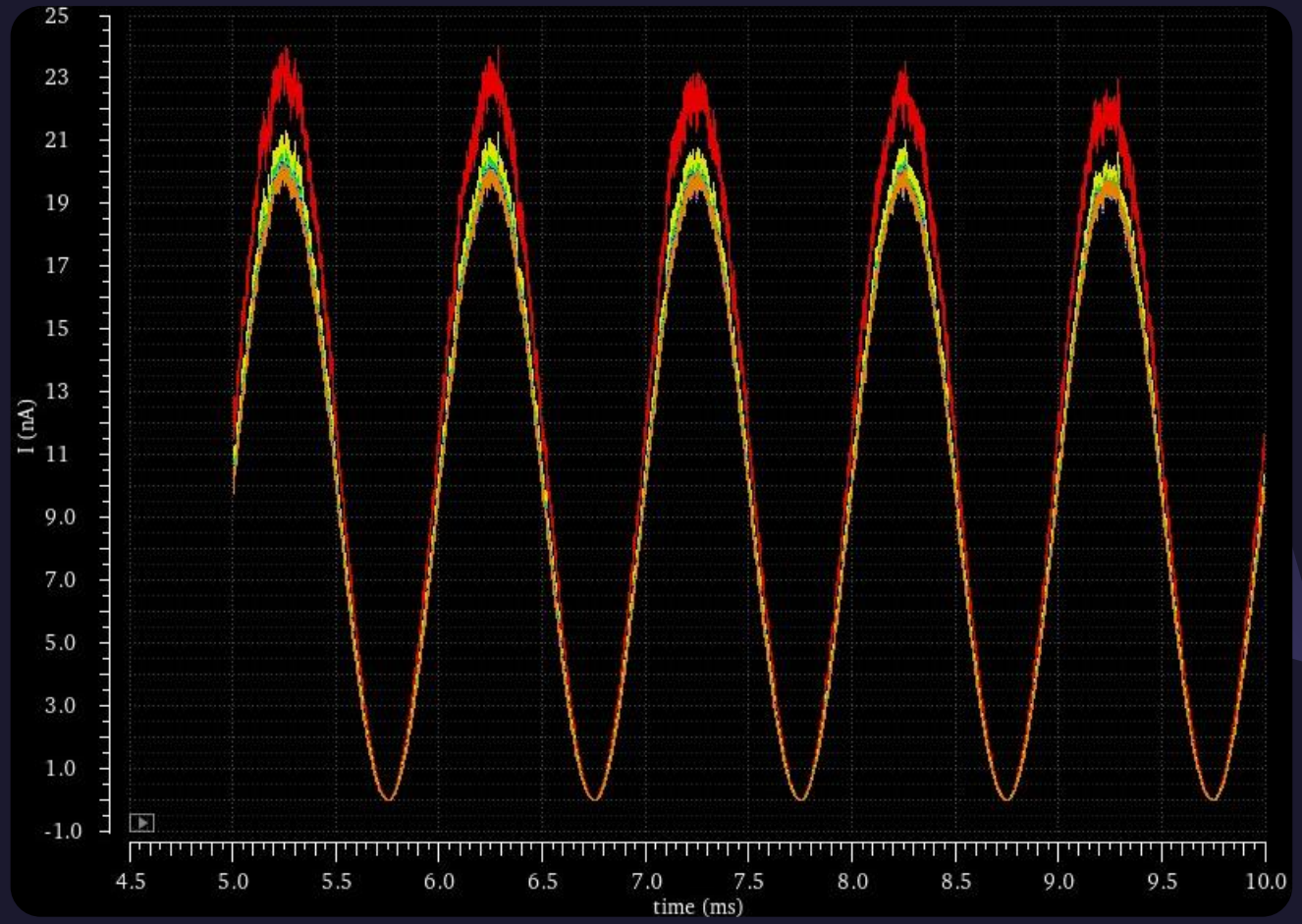
Dependence of ENOB on maximum input current amplitude

To verify that the approximated functions shown in the last slide are correct, we also plotted the same functions in Cadence Virtuoso.





Dependence of ENOB on aspect ratio of the n-mos



The first analysis is performed by keeping the Wn dimension fixed at 1µm and varying the Ln dimension from 1µm to 300nm (corresponding to an aspect ratio ranging from 1 to 3,33).

As we can see, the channel modulation effect becomes more evident as Ln decreases.

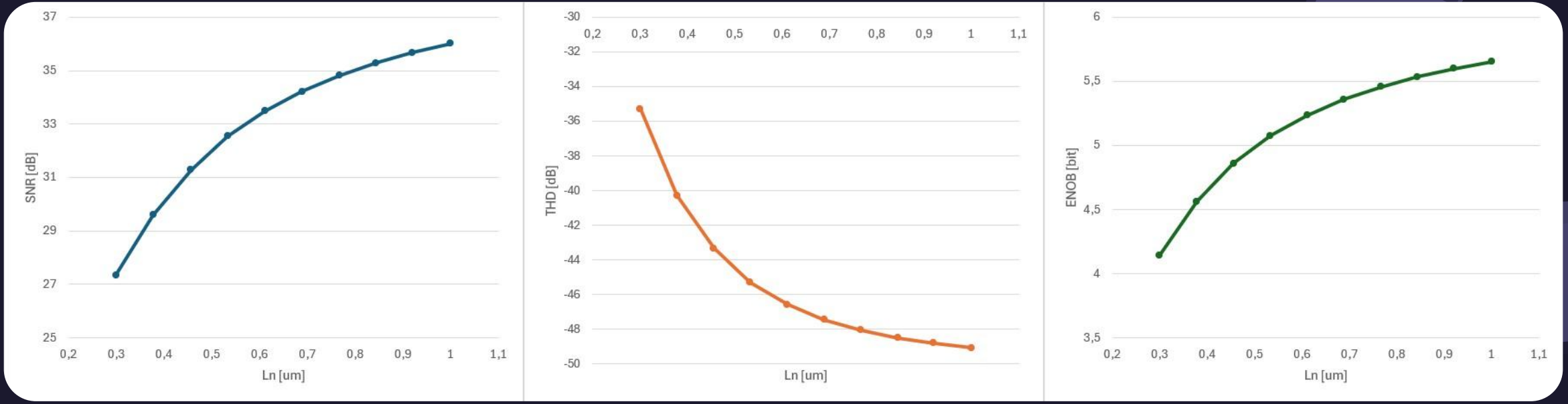
This introduce a distortion factor that increase as Ln decrease.

The simulation has been done with Vds=600mV and without charging the p-cap

Dependence of ENOB on aspect ratio of the n-mos

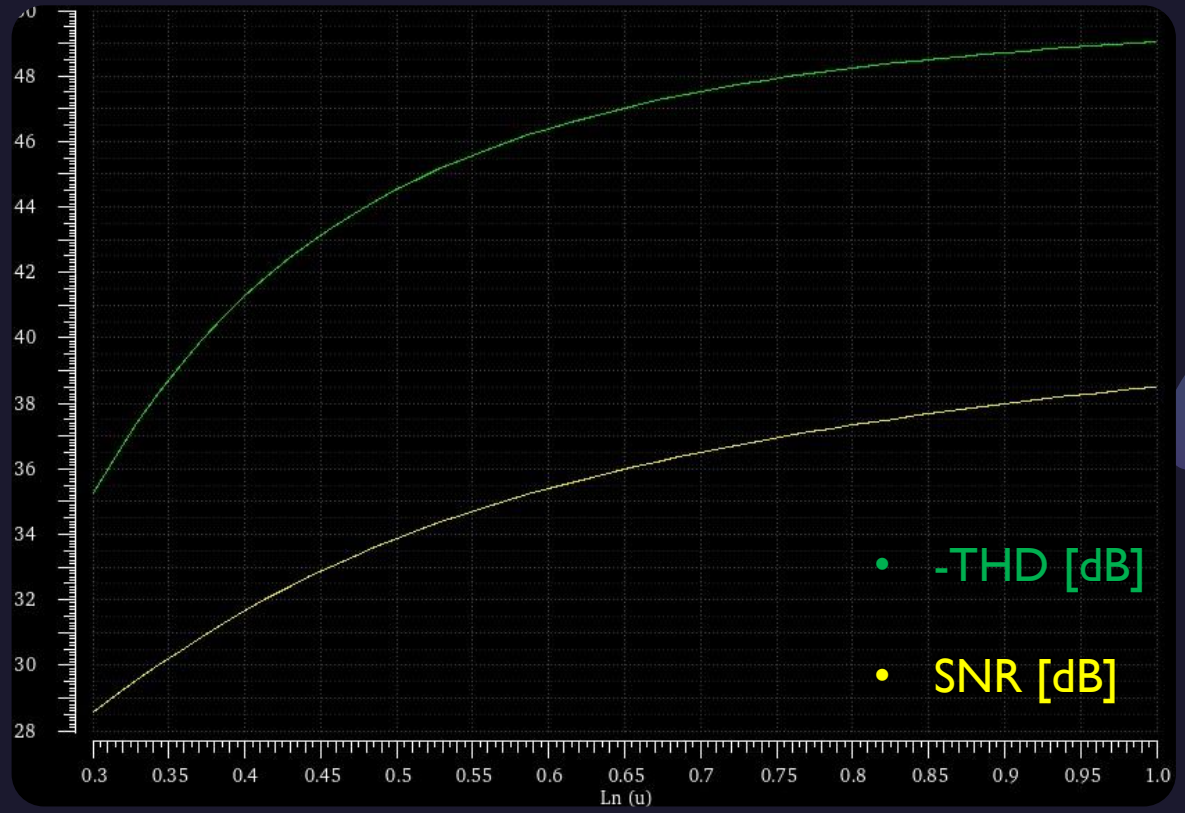
As we can see, increasing Ln value increase the $ENOB$ because it introduce less distortions due to channel modulation effect.

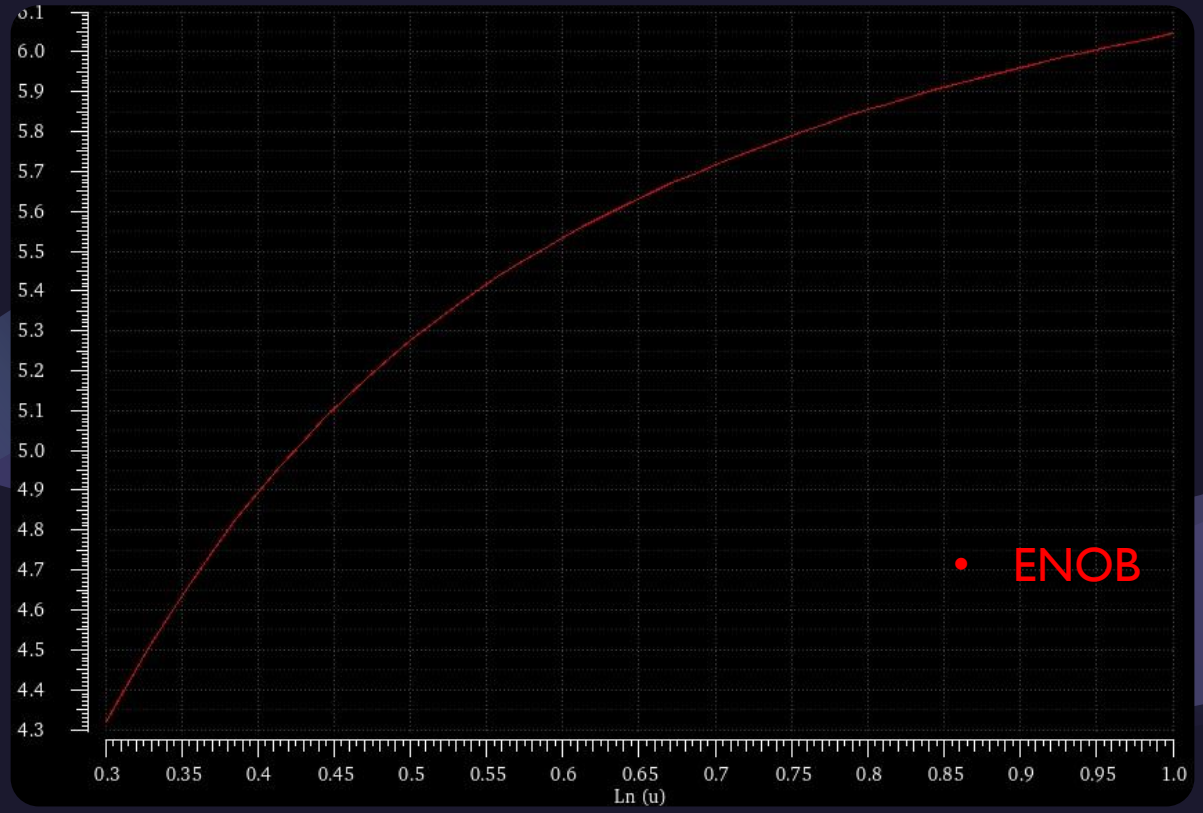
Moreover, increasing the n-mos dimensions also improves the SNR, because the noise in a mos transistor is inversely proportional to the device dimensions.



Dependence of ENOB on aspect ratio of the n-mos

To verify that the approximated functions shown in the last slide are correct, we also plotted the same functions in Cadence Virtuoso.

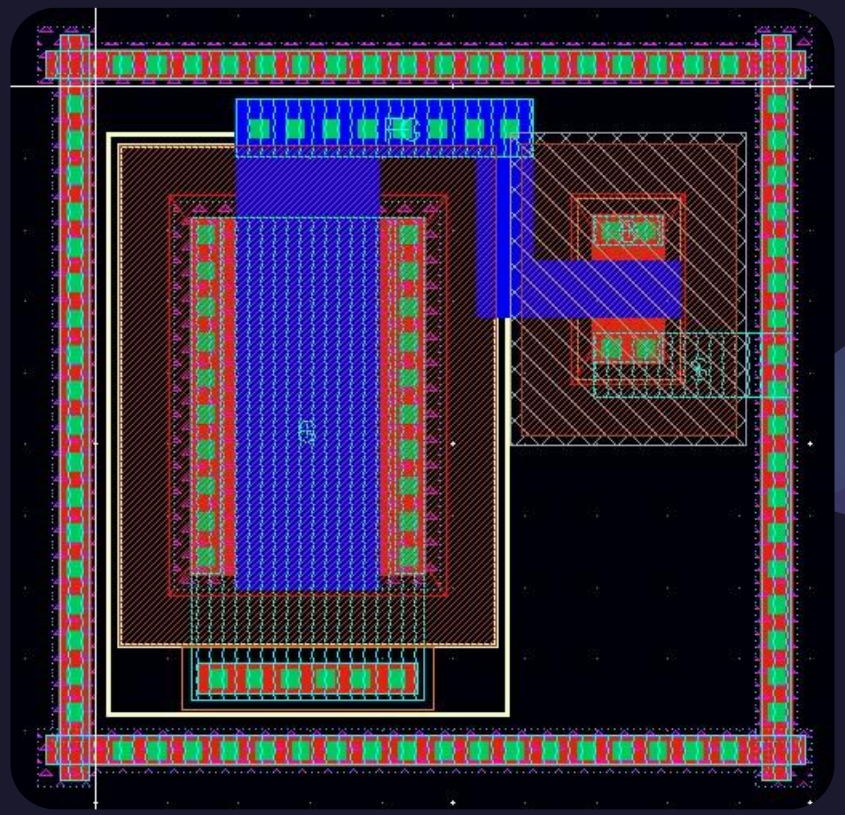




Floating gate cell layout

The floating gate cell designed have the following parameters:

- Ln = 800nm
- Wn = 1um
- Lcap = 2um
- Wcap = 5um



With this dimensions we can calculate the maximum drain current that ensure the mos to work in weak inversion.

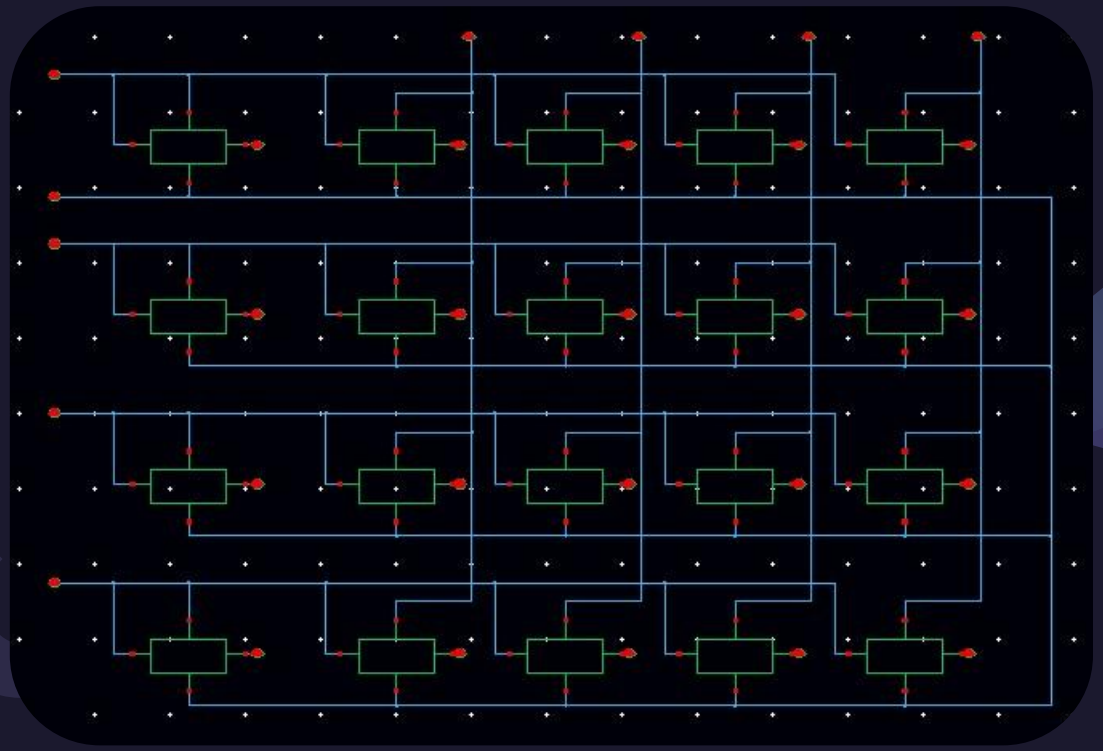
$$\frac{I_{ds}}{I_{dspec}} \leq \frac{1}{10} \rightarrow I_{ds} \leq \frac{I_{dspec}}{10} = \frac{I'_{dspec}\frac{W}{L}}{10} \cong 25nA$$

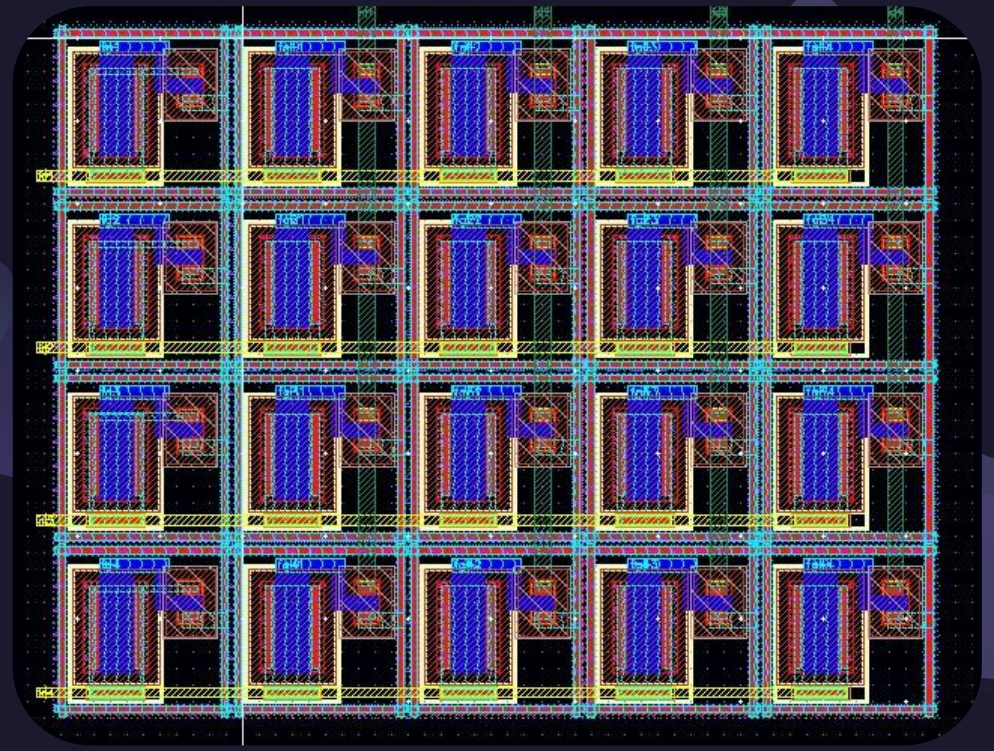
The designed cell can ensure an ENOB value greater than 5 bits with a Vds = 600mV and a maximum input current of 25nA.

4x4 vmm layout

The floating gate cell designed earlier can be connected to form a matrix of current mirrors.

In this case we designed a 4x4 vmm.





Thank you for your attention

